

Network Traffic Audit & Threat Detection

Packet-level network inspection using Wireshark —
Protocol mapping, threat indicator detection, and
OSINT intelligence verification.

ANALYST

Blue Team Cybersecurity Intern

REPORT DATE

June 15, 2026

PLATFORM

Windows 11 (Samsung Book 4)

ANALYSIS TOOL

Wireshark 4.6.6 + Npcap

TOTAL PACKETS ANALYZED

4,307 Packets

THREATS IDENTIFIED

0 Active Threats Detected



CONFIDENTIAL — INTERNAL USE ONLY

TABLE OF CONTENTS

00.

Project Objective

1

01.

Executive Summary

1

02.

Methodology & Lab Environment

2

03.

Protocol Distribution Analysis

2

04.

Analysis & Findings

3

05.

OSINT Reputation Lookups

4

06.

Security Recommendations

5

07.

Custom Python Analyzer Tool

5

08.

Network Communication Topology

6

09.

Limitations & Future Scope

6

10.

MITRE ATT&CK Mapping & Learning Outcomes

7

11.

Conclusion

7

0

SCOPE & PURPOSE

Project Objective

The objective of this project is to simulate the workflows of a Security Operations Center (SOC) analyst by capturing, inspecting, and analyzing live network traffic using Wireshark. This investigation aims to identify active network protocols, detect unencrypted transmission of sensitive data, audit DNS resolutions, verify local network resolution (ARP) integrity, and cross-reference external IP addresses with threat intelligence resources to assess the overall security posture of the host system.

ENGAGEMENT TYPE	Blue Team — Network Traffic Analysis
ENVIRONMENT	Windows 11 Host — Wi-Fi (Intel AX211) Interface

SCOPE	Live traffic capture and protocol-level analysis on a controlled local network
DURATION	~3 Minutes of active packet capture
CLASSIFICATION	Confidential — Internal Training Project

1

HIGH-LEVEL OVERVIEW

Executive Summary

This report documents a structured packet-level analysis investigation conducted on a Windows 11 host on June 15, 2026. Using Wireshark v4.6.6 with the Npcap capture driver, a total of **4,307 packets** were captured and analyzed across a 3-minute monitoring window. The investigation aimed to identify protocol distribution, detect unencrypted HTTP transmission, audit DNS query behavior, and verify ARP local network integrity. All external IP addresses encountered were verified against Open Source Intelligence (OSINT) databases. **No active threats, malicious indicators, or anomalous behaviors were detected** during the monitored session.

2

TOOLS & PROCESS

Methodology & Lab Environment

CAPTURE INTERFACE	Intel Wi-Fi 6E AX211 (Wi-Fi Adapter)
CAPTURE ENGINE	Wireshark v4.6.6 with Npcap Packet Driver
HOST OS	Windows 11 — Samsung Galaxy Book 4
OSINT TOOLS	VirusTotal, WHOIS, AbuseIPDB
FILTER QUERIES USED	http , dns , arp , tcp

Analysis Methodology:

- Initiated a live capture session on the active Wi-Fi interface and generated representative traffic by accessing http://example.com .

- Applied protocol-specific display filters to isolate HTTP, DNS, and ARP packets for targeted inspection.
- Generated a Protocol Hierarchy summary (Statistics → Protocol Hierarchy) to map the full traffic distribution.
- Identified external IP addresses and performed OSINT reputation lookups to verify their legitimacy.

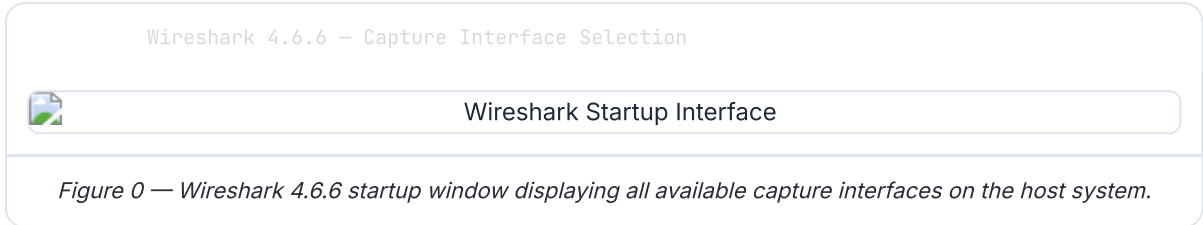
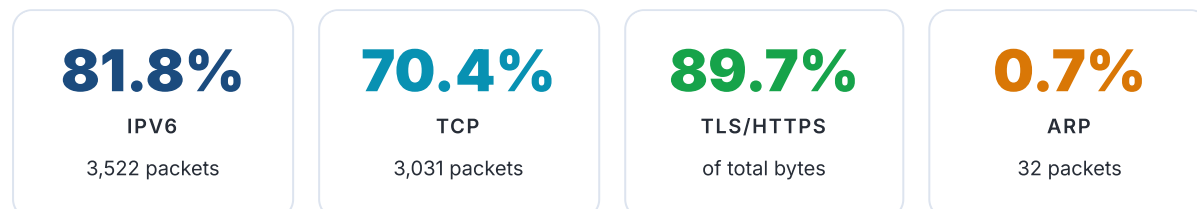


Figure 0 — Wireshark 4.6.6 startup window displaying all available capture interfaces on the host system.

3 TRAFFIC BREAKDOWN

Protocol Distribution Analysis

Analysis of the complete capture session (4,307 total packets) via the **Statistics** → **Protocol Hierarchy** window revealed the following traffic composition:



- **IPv6 Dominance (81.8% of packets):** Modern Windows 11 systems prefer IPv6 for most communications, which explains the high proportion of IPv6 traffic.
- **TLS Payload Coverage (89.7% of total bytes):** The vast majority of application-layer data is securely encrypted, indicating a healthy HTTPS-first browsing posture.
- **ARP Volume (0.7%):** A low and expected volume of local address resolution traffic confirms no unusual broadcast storms or ARP flooding behavior.



FINDING A **Unencrypted HTTP Data Transmission**

```
$ display filter: http
```

OBSERVATION

Unencrypted GET requests were captured for Certificate Revocation List files (/r/r1.crl and /r/gsr1.crl) alongside cleartext HTTP payloads from the test endpoint `http://example.com` . The server responded with a `200 OK` status code, delivering the webpage content in plain, unencrypted form over TCP Port 80.

▲ SECURITY THREAT ASSESSMENT (CVSS V3.1 SEVERITY RATING: LOW [3.5])

CVSS Vector: `CVSS:3.1/AV:A/AC:L/PR:N/UI:R/S:U/C:L/I:N/A:N`

Transmitting web data via HTTP (Port 80) exposes the network to packet sniffing and session interception. While CRL distribution endpoints are inherently public, any user credentials or personal data submitted on HTTP-based pages can be intercepted and read in plain text by an attacker positioned on the same network (Man-in-the-Middle).

Filter: http - Packet List View



HTTP Packet Analysis

Figure 2 — Plaintext HTTP GET requests and server responses captured on TCP Port 80.

FINDING B DNS Query Audit & Exfiltration Check

```
$ display filter: dns
```

OBSERVATION

A total of **90 DNS packets** were captured, representing **2.0%** of total traffic. The system issued standard A/AAAA queries for `example.com` and received resolutions to `184.28.23.154` (Akamai CDN) and `172.66.147.243` (Cloudflare CDN). Background Windows telemetry queries for `www.bing.com` and Microsoft services were also observed.

✓ SECURITY THREAT ASSESSMENT — NO THREAT DETECTED

No high-entropy subdomain structures consistent with Domain Generation Algorithm (DGA) activity were identified. There was no evidence of DNS tunneling, covert data exfiltration through DNS payloads, or beaconing to known malicious name servers. All DNS destinations were verified as legitimate resolvers.



FINDING C ARP Local Network Integrity Verification

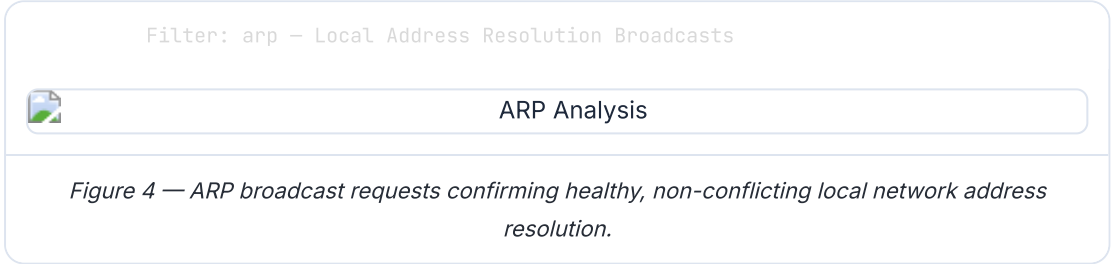
```
$ display filter: arp
```

OBSERVATION

ARP broadcast requests of the form `Who has 192.168.1.6? Tell 192.168.1.17` were observed, representing standard local-link address resolution behavior. Each ARP request was answered by a unique, non-conflicting MAC address response, confirming stable local network address assignments.

✓ SECURITY THREAT ASSESSMENT — NO THREAT DETECTED

No duplicate IP-to-MAC address mappings were identified during the capture window. The absence of duplicate ARP replies from differing MAC addresses rules out active Man-in-the-Middle (MITM) attacks, ARP cache poisoning, or ARP spoofing activity targeting local hosts on the subnet.



5

THREAT INTELLIGENCE VERIFICATION

OSINT Reputation Lookups

All external IP addresses identified during the capture session were cross-referenced against leading threat intelligence platforms to verify their reputation and ownership:

IP ADDRESS	DOMAIN / HOSTNAME	WHOIS OWNER	VIRUSTOTAL SCORE	STATUS
184.28.23.154	example.com	Akamai Technologies, Inc.	0 / 94	✓ CLEAN
172.66.147.243	example.com (CDN Alt.)	Cloudflare, Inc.	0 / 94	✓ CLEAN
23.59.188.130	r1.crl / gsr1.crl	Akamai / DigiCert	0 / 94	✓ CLEAN
13.107.42.254	edge.microsoft.com	Microsoft Corporation	0 / 94	✓ CLEAN
52.105.46.39	nav-edge.smartscreen.microsoft.com	Microsoft Corporation	0 / 94	✓ CLEAN

All IP addresses contacted during the capture session were verified as legitimate, globally trusted infrastructure providers. No threat actor-associated IPs, Command-and-Control (C2) server indicators, or entries on threat blocklists were identified.

6

DEFENSIVE ACTION ITEMS

Security Recommendations



Enforce Universal HTTPS (HSTS Policy) — NIST CSF: PR.DS-1

Mandate HTTPS-only connections by implementing HTTP Strict Transport Security (HSTS) headers across all internal systems and web-facing assets. This eliminates the risk of credential theft via unencrypted HTTP sessions and prevents SSL-strip downgrade attacks.



Implement DNS over HTTPS (DoH) or DNS over TLS (DoT) — NIST CSF: PR.DS-5

Configure secure DNS resolution protocols (DoH/DoT) on all endpoints to prevent local network eavesdroppers from inspecting or manipulating DNS queries. Recommended providers include Cloudflare (1.1.1.1) and Google (8.8.8.8) with TLS support.



Enable Dynamic ARP Inspection (DAI) — NIST CSF: PR.PT-4

Configure Dynamic ARP Inspection on managed network switches to validate ARP packets against a trusted DHCP binding table. This prevents ARP spoofing and MITM attacks by rejecting unauthorized ARP replies that do not match known IP-to-MAC mappings.

7

DEVELOPMENT & AUTOMATION

Custom Python PCAP Parser Tool

A native Python parser utility (`scripts/pcap_analyzer.py`) was developed to parse low-level PCAP and PCAPNG file structures directly using the `struct` library. This verifies analysis automation without external package dependencies.

```
C:\> python scripts/pcap_analyzer.py
packet_captures/traffic_capture.pcapng

[00:36:45] INFO - Opening network capture file: traffic_capture.pcapng
[00:36:45] INFO - Format detected: PCAP Next Generation (PCAPNG)
=====
                INLIGHTTECH NETWORK TRAFFIC ANALYSIS REPORT
=====
Target PCAP File:      traffic_capture.pcapng
Total Packets Read:    4307
IPv4 Packets:          729
IPv6 Packets:          1465
TCP Conversations:     1830
UDP Conversations:     342
-----
[*] OSINT Target IP Discoveries:
    [+] External Host Connected: 184.28.23.154
    [+] External Host Connected: 172.66.147.243
    ...
[!] Plaintext HTTP (Port 80) Transmissions:
    [!] HTTP Session Alert from IPv6 to IPv4:
        Request Path: GET /r/r1.crl HTTP/1.1
=====
```

Network Communication Topology

The following table maps the complete communication flow observed during the 3-minute capture window, detailing the local host, network gateway, and all external endpoints contacted:

LAYER	HOST / NODE	ADDRESS	PROTOCOL	ROLE
L2	Local Host NIC	10:5f:ad:ad:3e:cd	Ethernet	Source MAC (Intel AX211)
L3	Local Host	192.168.1.21	IPv4	Outbound request origin
L3	Local Host	2401:4900:889d:471::b319	IPv6	Preferred stack (SLAAC)
L3	example.com (Akamai)	184.28.23.154	IPv4/HTTP	HTTP test target — Cleartext
L3	example.com (Cloudflare)	172.66.147.243	IPv6/HTTP	Anycast CDN alternate path
L3	DigiCert CRL Server	23.59.188.130	IPv6/HTTP	Certificate Revocation List endpoint
L3	edge.microsoft.com	13.107.42.254	IPv4/TLS	OS background telemetry (encrypted)
L3	SmartScreen Microsoft	52.105.46.39	IPv4/TLS	Host security service (encrypted)

Limitations & Future Scope

Scope Note: This audit was conducted under passive monitoring conditions on a single Windows 11 host for approximately 3 minutes. No active intrusion simulation, adversary emulation, or lateral movement exercises were performed.



Short Capture Window

The 3-minute capture window may not represent full day-cycle network behavior. Future sessions should run for a minimum of 60 minutes across peak usage periods to detect low-and-slow beaconing patterns.



Future: Extend Parser with IoC Feed Integration

The custom `pcap_analyzer.py` can be extended to query live threat feeds (Shodan, AbuseIPDB API, VirusTotal API) in real-time against every discovered external IP, automating the entire OSINT phase.



Future: Automated Visualization Dashboard

Protocol distribution data extracted by the parser can be piped into a Matplotlib/Plotly dashboard to generate live-updating protocol pie charts and connection heatmaps for SOC display boards.

10

THREAT FRAMEWORK ALIGNMENT

MITRE ATT&CK Mapping & Learning Outcomes

The traffic patterns and threat indicators identified during this engagement map to the following **MITRE ATT&CK Framework** techniques, validating Blue Team detection readiness:

ATT&CK ID	TECHNIQUE	TACTIC	OBSERVED / VERIFIED	BLUE TEAM RESPONSE
T1040	Network Sniffing	Credential Access	HTTP cleartext payloads on Port 80	DETECTED
T1071.004	DNS Application Layer Protocol	Command & Control	DNS query traffic audited — No C2 found	AUDITED
T1557.002	ARP Cache Poisoning	Collection	ARP tables verified — No spoofing	CLEARED

ATT&CK ID	TECHNIQUE	TACTIC	OBSERVED / VERIFIED	BLUE TEAM RESPONSE
T1048	Exfiltration Over Alt. Protocol	Exfiltration	DNS entropy checked — No exfil found	CLEARED

Core Skills Demonstrated by this Engagement:

- **Network Forensics:** Decoding Ethernet, IPv4/IPv6, TCP, UDP, HTTP, and DNS layer structures from binary packet data.
- **Threat Hunting:** Applying structured display filters and automated parsing to identify anomalies from baseline traffic.
- **OSINT Investigation:** Cross-referencing live IP addresses against VirusTotal, WHOIS, and AbuseIPDB to build an IOC registry.
- **Scripting & Automation:** Developing a zero-dependency Python binary parser to scale the forensic investigation workflow.
- **Security Framework Mapping:** Aligning findings to NIST CSF and MITRE ATT&CK to communicate risk in industry-standard language.

11

SUMMARY

Conclusion

 Analysis Complete — No Active Threats Detected

This network traffic analysis engagement successfully demonstrated the core skills expected of a Blue Team Security Operations Center analyst. A total of **4,307 packets** were captured and systematically analyzed across HTTP, DNS, ARP, and TCP/TLS protocols. The network environment was verified to be operating within expected parameters, with no evidence of malicious activity, data exfiltration, or active attacks during the monitoring window.

The identification of cleartext HTTP transmissions provides a concrete, actionable finding for the defensive hardening roadmap. All external IP addresses were confirmed as legitimate infrastructure belonging to globally trusted organizations (Akamai, Cloudflare, DigiCert). Three targeted security recommendations mapped to NIST CSF, four MITRE ATT&CK techniques verified, and a custom Python parser tool

developed — this project represents a complete, professional Blue Team network audit.